

CO2

ASSESSMENT TOOL-3

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DATE:27.07.2026

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Monday

COURSE CODE : CSA1103

COURSE NAME : OOAD

1. Smart Pharmacy Management System – Use Case Diagram

(a) System Requirements

The Smart Pharmacy Management System should allow:

- Customers to search medicines.
- Customers to upload prescriptions.
- Customers to place medicine orders.
- Customers to make online payments.
- Pharmacists to verify prescriptions.
- Pharmacists to dispense medicines.
- Administrators to manage medicine inventory.
- Administrators to manage user accounts.

(b) Actors

The main actors are:

1. **Customer**
2. **Pharmacist**
3. **Administrator**
4. **Payment Gateway** – external system for online payment.

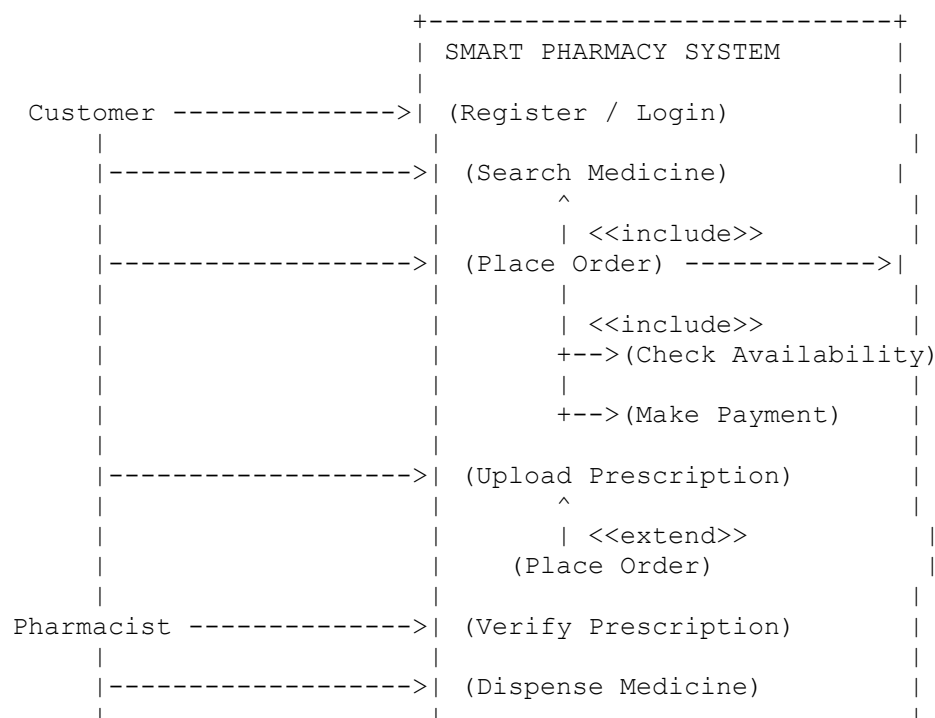
Main Use Cases

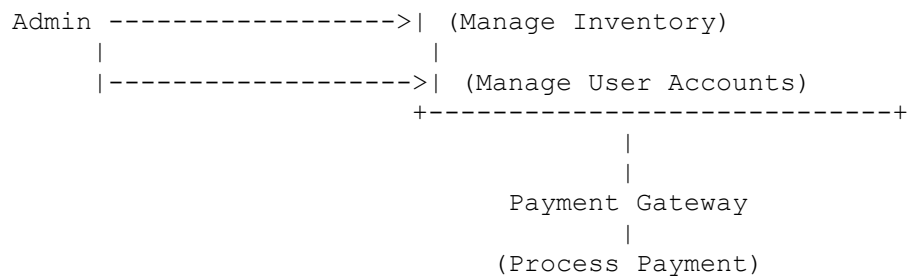
- Login / Register
- Search Medicine
- Upload Prescription
- Place Order
- Make Payment
- Verify Prescription
- Dispense Medicine
- Manage Inventory
- Manage User Accounts

(c) Use Case Relationships

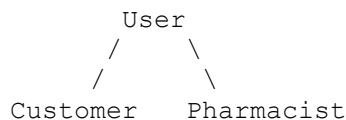
- **Place Order** <<include>> **Search Medicine**
- **Place Order** <<include>> **Check Medicine Availability**
- **Place Order** <<include>> **Make Payment**
- **Upload Prescription** <<extend>> **Place Order** because only prescription-required medicines need a prescription.
- **Customer** and **Pharmacist** can be generalized from **User**.
- **Online Payment** is connected to the external **Payment Gateway**.

Use Case Diagram





Generalization



Conclusion: The Use Case Diagram clearly identifies the responsibilities of customers, pharmacists, administrators, and external payment services.

2. Smart Hotel Management System – Class Diagram

(a) Required Classes

The main classes are:

- **Guest**
- **Room**
- **Reservation**
- **Payment**
- **Receptionist**

An additional parent class **Person** can be used to demonstrate **inheritance**.

(b) Attributes and Methods

Person

Attributes:

- personId
- name
- phone

Methods:

- login()
- logout()

Guest

Attributes:

- guestId
- email

Methods:

- searchRoom()
- makeReservation()
- cancelReservation()

Receptionist

Attributes:

- employeeId

Methods:

- checkIn()
- checkOut()
- manageReservation()

Room

Attributes:

- roomNo
- roomType
- price
- status

Methods:

- checkAvailability()
- updateStatus()

Reservation

Attributes:

- reservationId
- checkInDate
- checkOutDate
- status

Methods:

- createReservation()
- cancelReservation()

Payment

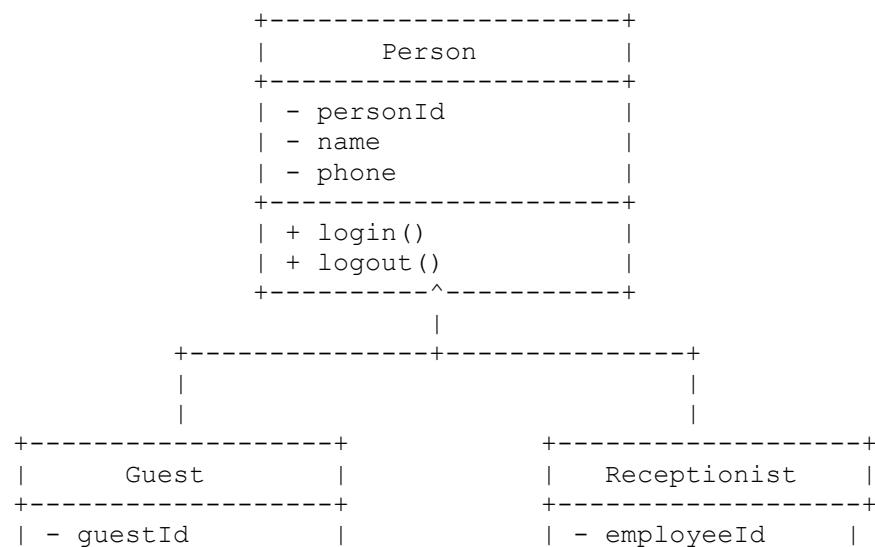
Attributes:

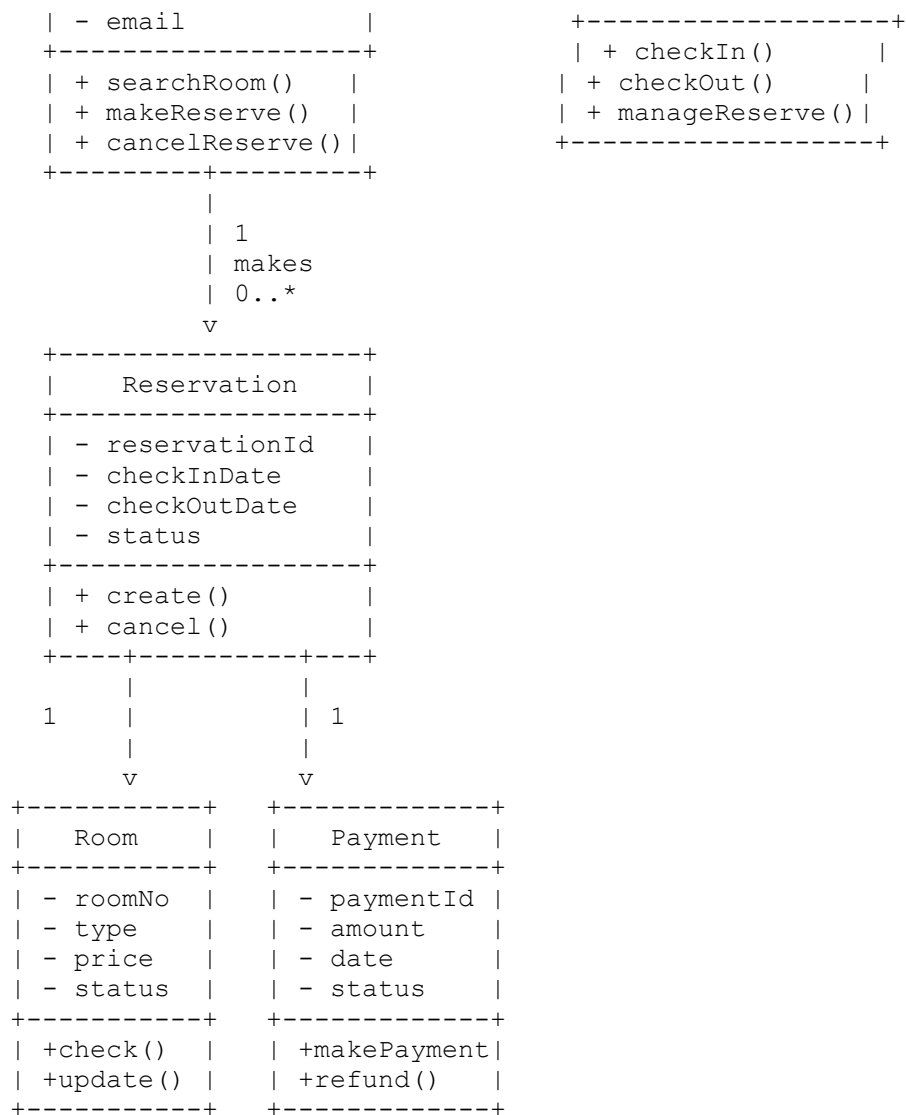
- paymentId
- amount
- paymentDate
- paymentStatus

Methods:

- makePayment()
- refundPayment()

(c) Class Diagram





Relationships

- **Inheritance:** Person → Guest, Receptionist
- **Association:** Guest makes Reservation.
- **Aggregation:** Hotel can have many Rooms, and rooms can exist independently.
- **Composition:** Reservation contains Payment details associated with that reservation.
- **Multiplicity:** One Guest can have 0..* Reservations; each Reservation is for 1 Room.

Conclusion: The class diagram defines the system's **structure, attributes, methods, relationships, and multiplicity**.

3. Online Banking System – Sequence Diagram

(a) Participating Objects

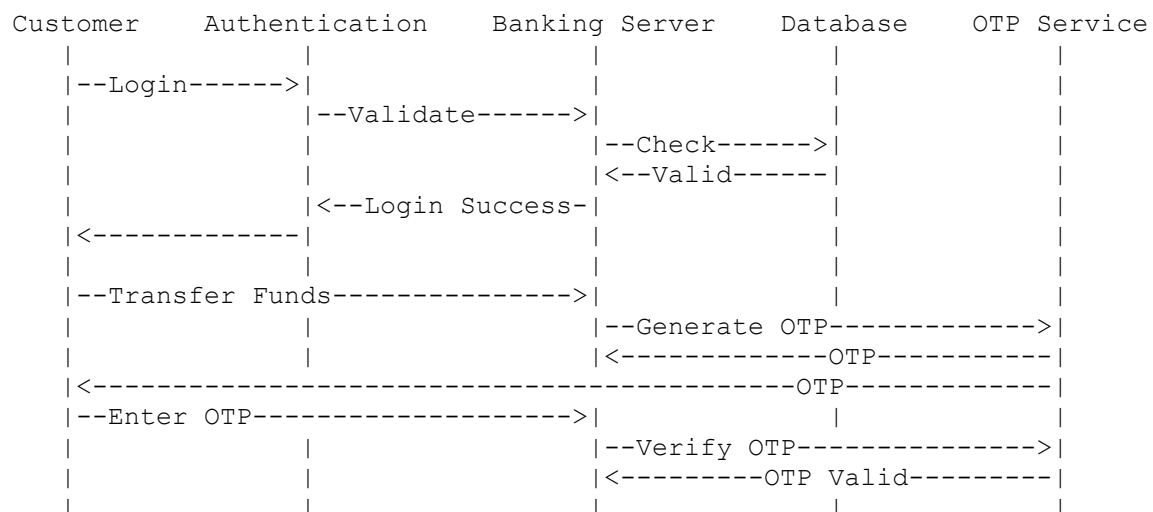
The main objects are:

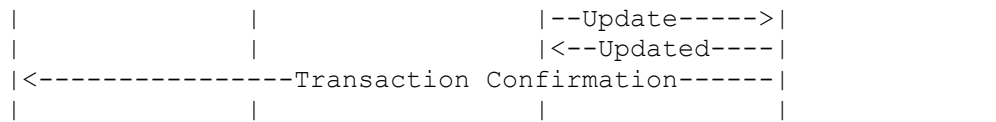
1. **Customer**
2. **Authentication Service**
3. **Banking Server**
4. **Database**
5. **OTP Service**

(b) Sequence of Operations

1. Customer enters login details.
2. Authentication Service validates credentials.
3. Banking Server checks customer information in the database.
4. Customer requests fund transfer.
5. Banking Server generates OTP.
6. OTP is sent to the customer.
7. Customer enters OTP.
8. OTP is verified.
9. Banking Server updates the account balance.
10. Database stores transaction details.
11. Customer receives confirmation.

(c) Sequence Diagram





Conclusion: The Sequence Diagram shows the **time-based order of communication** between the customer and system components during a secure fund transfer.

4. Passport Application System – Activity Diagram

(a) Workflow

The passport application process includes:

1. Applicant submits application.
2. Supporting documents are uploaded.
3. Identity verification is performed.
4. Processing fee is paid.
5. Application is reviewed.
6. Application is either approved or rejected.
7. If approved, passport is printed and delivered.

(b) Decision and Parallel Activities

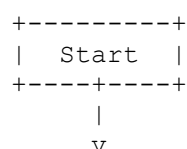
Decision Points

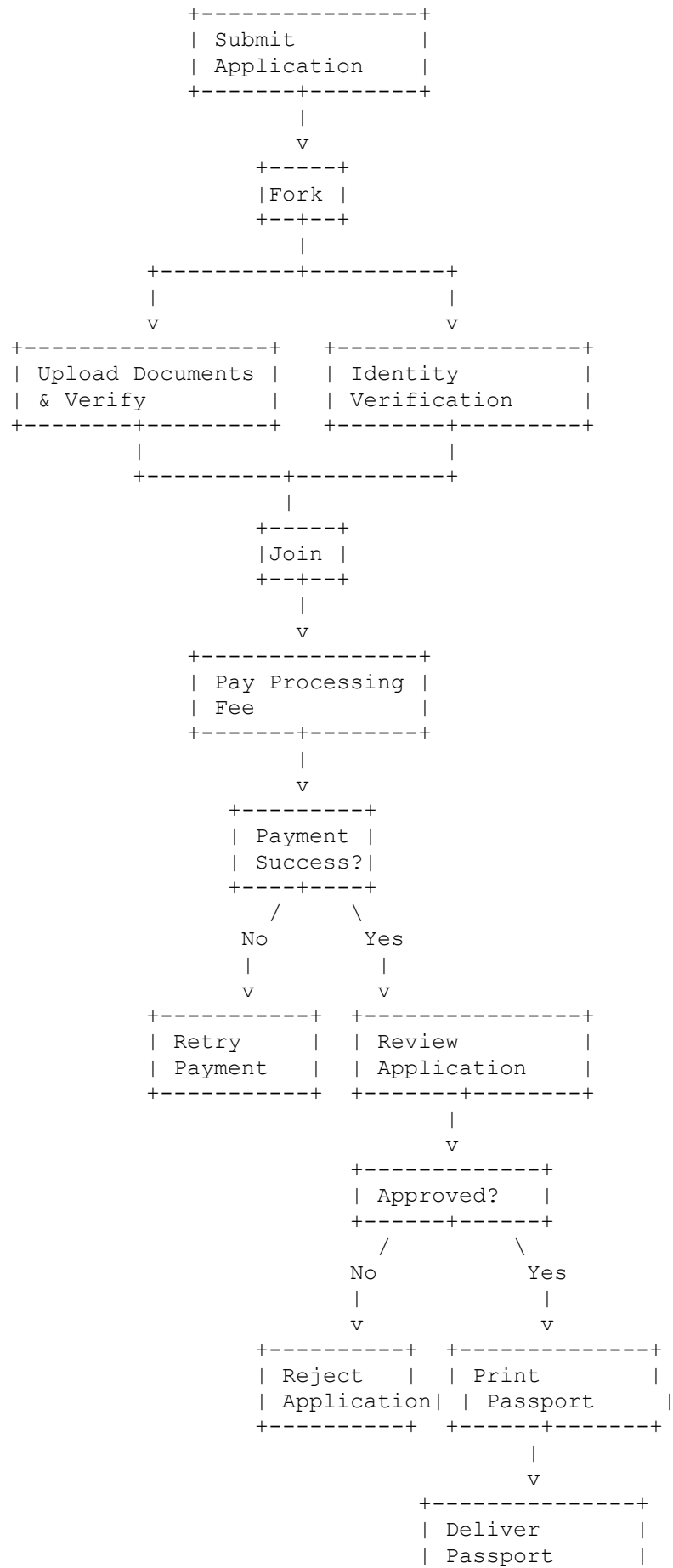
- Are documents valid?
- Is identity verified?
- Is payment successful?
- Is the application approved?

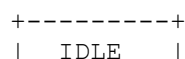
Parallel Activities

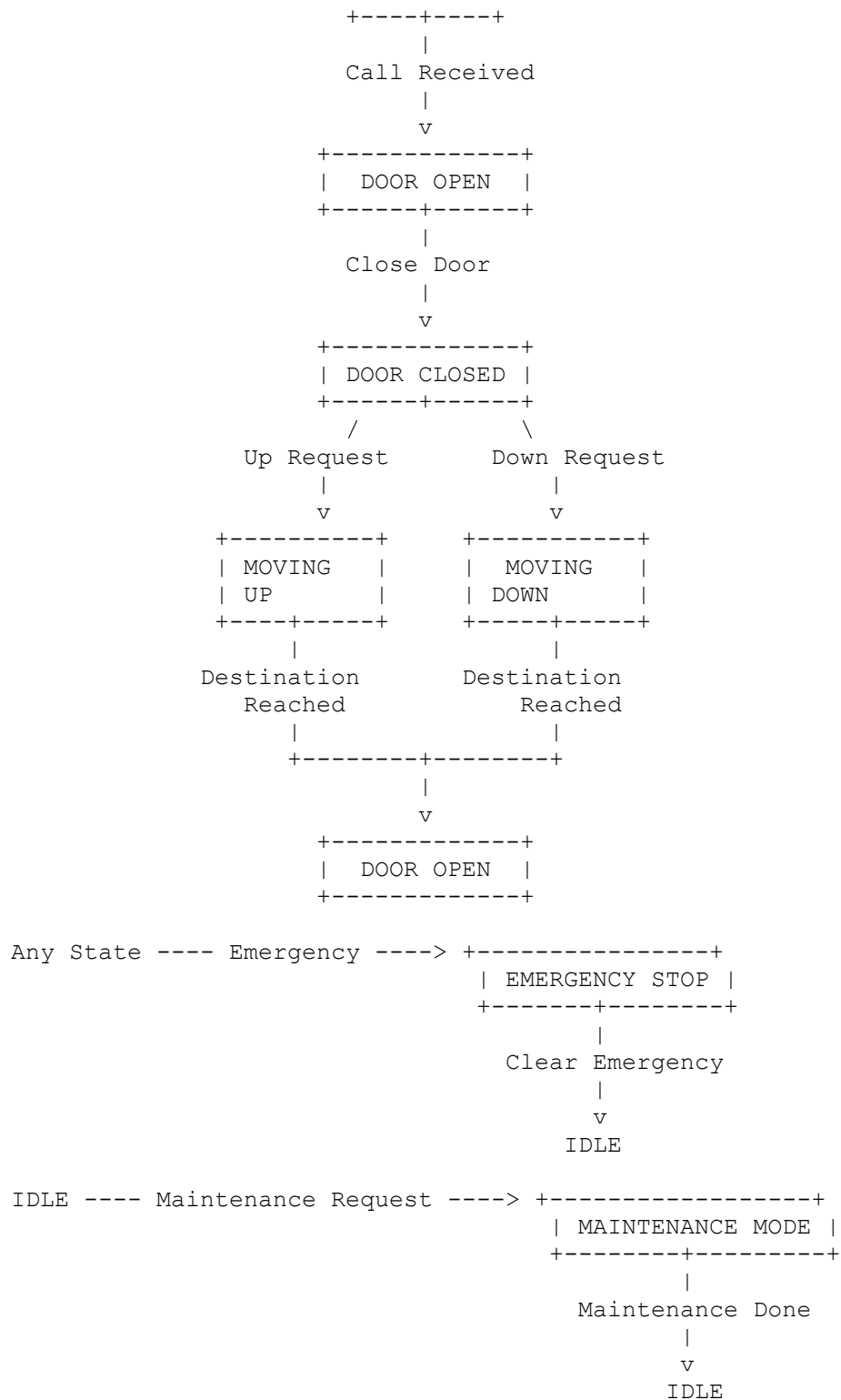
After submitting the application, **document verification** and **identity verification** can be performed concurrently.

(c) Activity Diagram









Conclusion: The State Diagram represents the elevator's behavior by showing **states, events, and transitions** under normal, emergency, and maintenance conditions.

6. Smart Home Automation System – Deployment Diagram

(a) Deployment Environment

The Smart Home system contains:

Hardware Nodes

- Smart Sensors
- Smart Lights and Appliances
- IoT Controller
- Home Gateway / Router
- Cloud Server
- User Smartphone

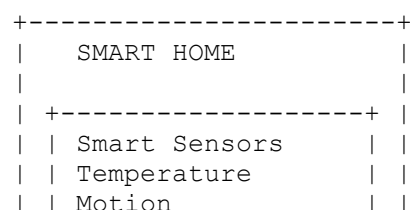
Software Artifacts

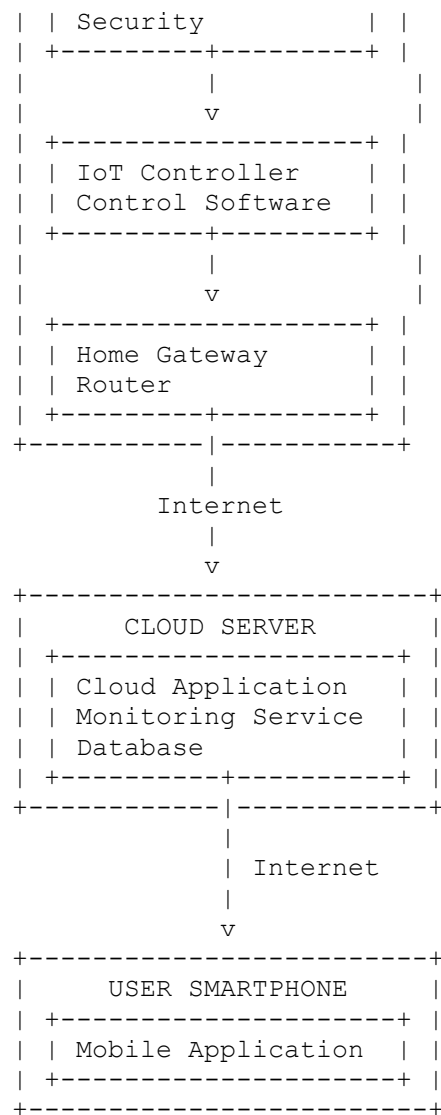
- Sensor Firmware
- IoT Control Software
- Home Automation Application
- Cloud Monitoring Service
- Mobile Application
- Database

(b) Communication

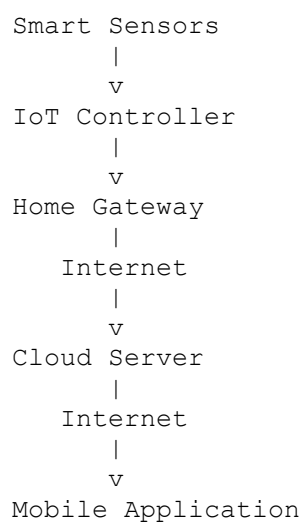
- Sensors communicate with the **IoT Controller**.
- IoT Controller communicates with the **Home Gateway**.
- Home Gateway connects to the **Cloud Server** through the Internet.
- User Mobile App communicates with the Cloud Server remotely.

(c) Deployment Diagram





Physical Architecture



Conclusion: The Deployment Diagram shows the **physical hardware nodes, software artifacts, deployment locations, and communication channels** of the Smart Home Automation System.